VAST-GWO: A Verified Archive-Guided Sparse-Evaluation Grey Wolf Optimizer for Global Optimization and 3D UAV Path Planning

Draft – `sparse_q` research branch

***Abstract*—Metaheuristic optimizers are increasingly deployed where each objective evaluation is expensive, so the real-evaluation budget—not arithmetic—is the resource that decides solution quality. The grey wolf optimizer (GWO) spends this resource uniformly over the whole pack and drives search from three leader wolves only, which invites premature convergence and makes each evaluation less informative than it could be. This paper proposes VAST-GWO, a verified archive-guided sparse-evaluation GWO. In every generation, 100 cheap candidates are generated from three complementary routes: standard GWO leader moves (50), success-scale shells around the current alpha wolf (12), and multi-region shells and two-centre segments produced from a verified elite archive (38). Only $K = 10$ uniformly sampled candidates receive a real evaluation, and unevaluated candidates never influence leaders, archives or rankings—a verified-state discipline that keeps every piece of search memory backed by a billed evaluation. On CEC2017 at $D = 50$ with 100,000 true evaluations and 30 independent runs, VAST-GWO ranks first among seven methods (mean rank 2.24), significantly outperforming the 2026 hierarchical HAGWO, PSO, CLPSO, the 2026 adaptive oppositional AOGWO and standard GWO, and reaching statistical parity with the strong improved GWO. A controlled equal-slot ablation shows that the verified archive carries the improvement—removing it degrades the method to 27 losses in 29 functions, and replacing it with a single alpha-centred sampler collapses it to 2 wins—while the alpha-shell route is optional once the archive is present. In a 209-dimensional, DEM-based 3D UAV delivery application with hard time windows, VAST-GWO is the only method that finds feasible routes in all 150 runs, attains the smallest censored mean time to feasibility, and produces the best final route quality against every baseline after Holm correction. These results indicate that, under limited real-evaluation budgets, the structure of candidate sources matters more than the complexity of the evaluation scheduler.**

***Index Terms*—Grey wolf optimizer; sparse evaluation; verified elite archive; multi-region search; limited evaluation budget; UAV path planning; CEC2017.**

I. Introduction

Nature-inspired optimizers are now a standard tool for engineering design, scheduling and path planning, because they make few assumptions about the objective and scale to black-box simulators. Their practical cost, however, is dominated by the objective function itself: in finite-element calibration, computational-fluid design or UAV mission planning a single evaluation may take seconds to hours, so the total number of *real* evaluations is the binding constraint rather than the number of arithmetic operations [1], [2], [3]. Under such budgets, an optimizer must answer one question well: *where should the next real evaluation be spent?*

The grey wolf optimizer (GWO) [4] is a particularly attractive starting point: it is parameter-light, easy to implement and has been applied across IoT, feature selection, agriculture and UAV planning [5], [6]. Its classic form, however, evaluates the entire pack every generation and derives all movement from the three leaders (α, β, δ). When evaluations are scarce this design has two consequences: budgets are consumed on candidates whose information value is unknown, and the search distribution concentrates around the same leaders, which is the classical route to premature convergence and stagnation on multimodal landscapes.

A large recent literature improves GWO from different angles. Operator-level variants add differential evolution and elimination [7], mutation with evolutionary population dynamics [8], teaching–learning hybrids [9], beetle-antennae hybrids [10], grey wolf–particle swarm hybrids [11], self-repulsion for feature selection [12] and improved engineering-oriented forms [13]. Learning-driven variants let reinforcement learning or Q-learning choose update strategies or leaders [14], [15], [16], [17], and recent drones-specific versions target UAV trajectory planning [18], [19]. These methods deliver strong results, but they are typically tuned and compared under protocols in which every individual is evaluated each generation or the budget is defined in iterations. They also tend to stack mechanisms, so it is rarely isolated *which* design element produces the gain.

This paper takes the evaluation budget itself as the design constraint and asks a deliberately narrow question: *given a fixed number of real evaluations per generation, how should the candidate pool be composed?* We propose VAST-GWO (Verified Archive-guided Sparse-evaluation Technique for Grey Wolf Optimization), which separates cheap candidate generation from real evaluation and admits information into the search state only after it has been verified by a billed evaluation. The candidate pool combines three routes—standard GWO moves, success-scale alpha shells, and multi-region candidates drawn from a verified elite archive—and only a small random subset of the pool is ever evaluated, which makes the composition of the pool the decisive design variable.

The contributions of this paper are summarized as follows.

- **A verified-state sparse-evaluation framework.** Each generation generates $M = 100$ cheap candidates, eval-

All experiments in this manuscript come from the frozen batch recorded in the accompanying repository; every reported number is reproducible from the archived runs, and the evaluation counts of all runs are audited by an independent probe.

uates $K = 10$ of them under a strictly audited true-evaluation budget, and lets no unevaluated candidate influence leaders, archives, rankings or statistics.

- **A verified multi-region archive as a candidate source.** The archive stores only real-evaluated improvements, enforces a separation threshold, and contributes shells around its centres and segments between them. A controlled equal-slot ablation shows that this route is the main performance engine: disabling the auxiliary routes reduces the method to 27 losses in 29 functions against the full version, and replacing the archive with a single alpha-centred sampler collapses it to 2 wins in 29 functions.
- **Complete evidence under one protocol.** A seven-method CEC2017 comparison at $D = 50$ (29 functions, 30 seeds, 100,000 true evaluations), a five-arm equal-slot source ablation, stagnation and early-budget probes, and two additional resource tiers ($D = 100$; 20,000 evaluations) are reported with identical initial populations and probe-audited budgets.
- **A demanding application.** On a 209-dimensional, DEM-based 3D UAV delivery problem with hard time windows on five real terrain scenarios, VAST-GWO is the only method that finds feasible routes in all 150 runs, has the smallest censored mean time to feasibility, and delivers the best final route quality against every baseline after Holm correction.
- **Design-space boundaries.** Directed screening, fixed exploration quotas, three Q-learning quota schedulers, stagnation radius boosting and success-direction steering are all tested with matched controls; none improves on uniform random allocation, which shows that the gains come from the candidate-source structure rather than from added scheduling complexity.

The remainder of this paper is organized as follows. Section II reviews related work. Section III presents VAST-GWO. Section IV describes the experimental protocol, Section V reports the optimization results, Section VI the ablation and mechanism analysis, and Section VII the UAV application. Section VIII discusses implications and scope, and Section IX concludes.

II. Related Work

A. Grey wolf optimizer and its variants

GWO [4] models the leadership hierarchy and cooperative hunting of grey wolves: the three best solutions guide the pack through encircling and attacking equations whose exploration–exploitation balance is controlled by a linearly decreasing parameter. Comprehensive reviews document its variants and applications in IoT and beyond [5], [6]. Representative recent improvements include differential-evolution and elimination mechanisms [7], mutation with evolutionary population dynamics and nonlinear population reduction [8], teaching–learning hybridization [9], beetle-antennae search hybridization [10], grey wolf–particle swarm hybrids [11], self-repulsion strategies [12], and engineering-oriented improved variants [13], as well as improved GWO for general engineering design [20]. The strong improved GWO [20] is used as a baseline in this paper. Learning-driven variants couple GWO with reinforcement learning or Q-learning for strategy, leader or quota selection [14], [15], [16], [17], and UAV-specific versions embed Q-learning or reinforcement learning into trajectory planning [18], [19]. The present work differs in that it neither adds a new update equation nor a learning scheduler; it reorganizes *which candidates exist* under a fixed number of real evaluations per generation.

B. Memory and archive mechanisms

Memory is a recurring idea in swarm intelligence, where it stabilizes search and preserves diversity. Recent GWO variants use elite memories, cross-search memory or two-archive structures to handle large-scale or multi-objective problems [21], [22]. VAST-GWO shares the motivation but differs in two respects. First, the archive is *verified*: only solutions that received a real evaluation may enter, so the archive never stores a guess. Second, the archive is not only a diversity reservoir—its centres and the segments between them directly *generate* candidates, which is the property that the ablation of Section VI isolates.

C. Optimization under expensive evaluations

When evaluations are expensive, surrogate-assisted evolutionary algorithms are the dominant paradigm: models of the objective guide the search and are updated with a limited number of real evaluations [1], [2], [3]. VAST-GWO is complementary rather than competing: it uses no surrogate, and it targets the regime where each real evaluation is affordable but scarce enough that the per-generation allocation must be fixed and audited. This makes the comparison clean—every method receives exactly the same number of billed evaluations—and it isolates candidate-source design from modelling effort.

III. Proposed Method

A. Overview and verified-state discipline

Fig. 1 shows one generation of VAST-GWO. A candidate pool of $M = 100$ positions is generated from three routes without any objective evaluation. Exactly $K = 10$ pool members are then sampled uniformly at random and evaluated with the real objective; each call is billed, and a runtime probe asserts that the audited count equals the budget exactly. The evaluated subset drives all state updates: the leaders α, β, δ, the verified elite archive, and the statistics used by the routes. Unevaluated candidates have no effect on any of them—the verified-state discipline. The budget of 100,000 evaluations decomposes as 50 initialization evaluations plus 9,995 generations of ten evaluations.

B. Candidate generation routes

a) G1: GWO leader moves (50 candidates).: For each wolf i the standard aggregation of the three leader-attracted positions is formed,

$$X_i^{G1} = \tfrac{1}{3} \sum_{k \in \{\alpha,\beta,\delta\}} \big(X_k - A_k \odot |C_k \odot X_k - X_i|\big), \qquad (1)$$

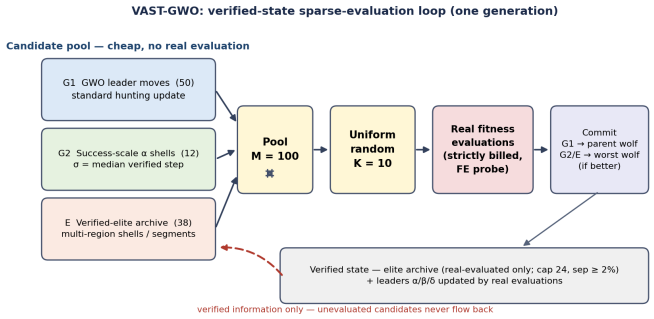


Fig. 1. One generation of VAST-GWO. A pool of 100 cheap candidates is generated from three routes; only $K = 10$ uniformly sampled candidates receive a real (billed) evaluation, and only verified information flows back into the leaders and the archive.

with $A_k = 2a\,r_1 - a$, $C_k = 2r_2$, $r_1, r_2 \sim U(0,1)^D$, and the convergence factor $a = 2 - 2\,\mathrm{FE}/\mathrm{FE}_{\max}$ driven by the true evaluation count. This route preserves the classical hunting behaviour.

b) G2: success-scale alpha shells (12 candidates).: Local refinement is generated in shells around the incumbent alpha wolf,

$$X^{G2} = X_\alpha + \sigma \odot (2u-1), \qquad \sigma = \max\big(\mathrm{median}(\mathcal{S}_{25}),\ 0.01\,d_{\mathrm{diag}}\big), \tag{2}$$

where $\mathcal{S}_{25}$ is the set of the 25 most recent *verified* successful step lengths (distances of accepted moves) and d_{diag} is the diagonal of the search space. The scale therefore adapts to the observed geometry of successful moves instead of a fixed schedule.

c) E: verified elite archive (38 candidates).: The archive stores up to 24 real-evaluated solutions whose objective values improve the current worst archive member and whose distance to the nearest stored solution exceeds 2% of d_{diag}; each entry carries a credit that counts how often it has contributed to an accepted candidate. Nineteen candidates are single-centre shells around credit-weighted archive centres c,

$$X^{E_1} = c + \sigma\,(0.5 + v) \odot (2u - 1), \tag{3}$$

and nineteen are two-centre segments between archive centres c_1, c_2,

$$X^{E_2} = c_1 + \lambda\,(c_2 - c_1) + 0.25\,\sigma \odot (2u-1), \quad \lambda \sim U(0.15, 0.85). \tag{4}$$

Because archived centres are separated by construction, this route proposes candidates in several distinct regions of the search space simultaneously—the mechanism that the ablation in Section VI identifies as the source of the method's advantage.

C. Sparse allocation and commit rules

The $K = 10$ evaluated candidates are drawn uniformly at random from the pool without replacement, which keeps the allocation free of additional hyperparameters and, as Section VI shows, is not improved upon by screening, quota or learning-based alternatives. Commits follow the origin of the candidate: an evaluated G1 candidate replaces its parent wolf if it improves that parent's objective; an evaluated G2 or E candidate replaces the current worst wolf if it is strictly better. All accepted moves are appended to the verified-step record that drives σ, and the archive and leaders are updated exclusively from these evaluated results.

Algorithm 1 VAST-GWO (one run)

initialize N wolves by real evaluation; set α, β, δ and $\mathcal{A} \leftarrow \emptyset$; $\mathrm{FE} \leftarrow N$
while $\mathrm{FE} < \mathrm{FE}_{\max}$ **do**
  build pool: 50 G1 moves (Eq. 1), 12 G2 shells (Eq. 2), 38 archive candidates (Eqs. 3–4)
  sample $K = 10$ candidates uniformly without replacement
  for each sampled candidate x **do**
    $f(x) \leftarrow$ real evaluation; $\mathrm{FE} \mathrel{+}= 1$
    commit x according to its route (parent or worst wolf, if better)
    update verified-step record, archive and α, β, δ from verified results only
  end for
end while
return best verified solution

D. Algorithm and complexity

Algorithm 1 summarizes the procedure. Per generation the method performs $M = 100$ cheap candidate constructions of cost $O(MD)$ and $K = 10$ objective evaluations; the archive and shell statistics add $O(|\mathcal{A}|D)$ with $|\mathcal{A}| \le 24$. Relative to standard GWO, which evaluates $N = 50$ candidates per generation, the arithmetic overhead of the pool is modest while the number of billed evaluations per generation drops by a factor of five; the wall-clock comparison at equal budgets is reported in Section V-D.

IV. Experimental Setup

a) Benchmark and protocol.: All optimization experiments use the CEC2017 benchmark suite with F1 and F3–F30 (29 functions; the official package omits F2), dimension $D = 50$, 100,000 true evaluations, and 30 independent runs (seeds 3001–3030). Every algorithm receives the same initial population for each (function, seed) pair; the initialization random stream and the search random stream are separated. All batches run on MATLAB with exactly four parallel workers, atomic per-run storage and resumable execution, and an independent probe asserts that each run consumes exactly its budget [23]. An out-of-sample seed block (3071–3100) is used only for the ablation study.

b) Baselines.: The comparison set contains six methods, all implemented or audited in the same codebase: standard GWO [4]; the strong improved GWO [20]; HAGWO [24]; classic PSO with linearly decreasing inertia [25], [26]; CLPSO [27]; and AOGWO [16]. Baseline-specific parameters are summarized in Table I.

TABLE I
BASELINE CONFIGURATIONS (IDENTICAL BUDGET FOR ALL METHODS).

Method	Configuration
GWO	$N = 50$
I-GWO	$N = 50$, as published
HAGWO	$N = 50$, as published
PSO	$N = 50$, w: 0.9→0.4, $c_1 = c_2 = 2$, $v_{\max} = 0.2\,r$
CLPSO	$N = 50$, P_c schedule and refresh as published
AOGWO	$N = 50$, as published

TABLE II
FINAL SEVEN-METHOD COMPARISON (CEC2017, $D = 50$, 100,000 TRUE EVALUATIONS, 29 FUNCTIONS, 30 RUNS, IDENTICAL SEEDS). "MED. RATIO" IS THE MEDIAN PAIRED ERROR RATIO OF VAST-GWO OVER EACH BASELINE (BELOW ONE FAVOURS VAST-GWO); HOLM COUNTS PER-FUNCTION SIGNIFICANT WINS/LOSSES AFTER HOLM CORRECTION.

Method	Med. ratio	W-L	Holm $+/-$	Pooled p	Mean rank
VAST-GWO	–	–	–	–	**2.24**
I-GWO	0.938	17-12	$+12/-6$	0.55	2.28
HAGWO (2026)	0.806	22-7	$+19/-3$	2.2×10^{-36}	3.41
PSO	0.652	23-6	$+19/-2$	5.8×10^{-31}	4.41
CLPSO	0.668	21-8	$+20/-5$	7.4×10^{-27}	4.48
AOGWO (2026)	0.586	27-2	$+21/-0$	1.3×10^{-85}	5.07
GWO	0.525	28-1	$+24/-0$	6.7×10^{-103}	6.10

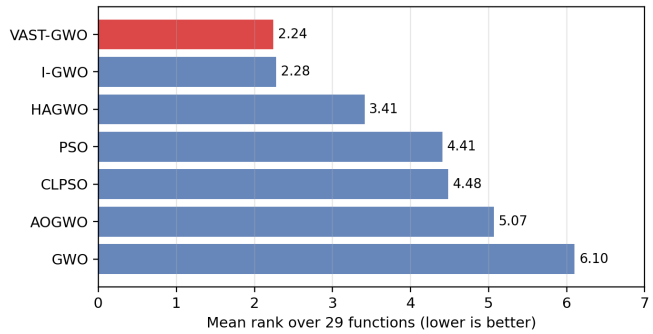


Fig. 2. Mean ranks over the 29 CEC2017 functions for the final seven-method comparison ($D = 50$, 100,000 true evaluations, 30 runs each).

TABLE III
BEHAVIOUR METRICS ON THE FORMAL TIER (PROBE-INSTRUMENTED METHODS).

Method	Recovery prob.	Long-stag. share	Useful novelty	Nov. rate Q4	Impr. rate Q4
S-star	0.823	0.813	0.00640	0.712	0.00308
S1-pool38	0.816	0.812	0.00590	0.695	0.00266
S3p-q25	0.808	0.821	0.00563	0.745	0.00272
S5p-q	0.780	0.835	0.00629	0.716	0.00264
B0-ANGWO	0.768	0.831	0.00578	0.690	0.00447
GWO	0.757	0.844	0.00347	0.588	0.00481

c) Statistics.: For every baseline, per-function paired Wilcoxon tests on identical seeds are reported together with Holm correction, the median paired error ratio (baseline over VAST-GWO), function-mean win–loss counts and the mean rank over the 29 functions. Pooled paired tests are reported for completeness but are not used as primary evidence. The same protocol applies to the ablation and to the application study.

V. RESULTS

A. Comparison with strong baselines

Table II reports the final seven-method comparison. VAST-GWO attains mean rank 2.24—first among the seven methods—ahead of the strong improved GWO (2.28), the 2026 hierarchical HAGWO (3.41), PSO (4.41), CLPSO (4.48), the 2026 adaptive oppositional AOGWO (5.07) and standard GWO (6.10). Against standard GWO the improvement is decisive: 28 wins in 29 functions, median ratio 0.525, Holm $+24/-0$. Against HAGWO, PSO, CLPSO and AOGWO the margins are statistically significant (see Table II), and against the strong improved GWO the comparison is a statistical tie (ratio 0.938, 17–12, pooled $p = 0.55$) with large function-level swings in both directions (e.g. $7\times$ better on F1 and $140\times$ on F9). The claim this comparison supports is therefore precise: VAST-GWO is the best-ranked method in the set and is significantly better than five of six baselines, while remaining statistically indistinguishable from the strongest classical improvement.

B. Early-budget behaviour and exploration

The advantage appears early and grows with the budget. At 10%, 20% and 50% of the evaluation budget the median error ratios against standard GWO are 0.802, 0.777 and 0.726, with 25, 26 and 27 wins out of 29 functions, respectively. Against the frozen full-evaluation ANGWO reference the ratios are 0.732 and 0.768 at 10% and 20%, and the simple-multimodal subset (F4–F10) reaches a ratio of 0.714. The probe-instrumented behaviour table (Table III) shows the same picture from the search dynamics: VAST-GWO recovers from stagnation windows more often than GWO (0.823 vs. 0.757) and spends less of the budget in long stagnation (0.813 vs. 0.844); its useful-novelty rate—evaluations that are both novel and convert into improvements—is 0.0064 versus 0.0035 for GWO. The improving-evaluation rate itself is lower (0.0031 vs. 0.0048), consistent with a search that places fewer, better-targeted evaluations.

C. Scalability and resource-constrained tiers

Two additional tiers confirm that the advantage is not tied to one operating point. At $D = 100$ (29 functions, ten seeds, 100,000 evaluations) VAST-GWO again ranks first (mean rank 1.72 versus 2.00, 2.33, 3.52 and 3.67 for the compared variants), with a 0.703 median ratio and 10 wins against 2 losses (Holm $+5/-1$) on the 12-function subset shared with the improved GWO. Under a five-times-smaller budget (20,000 evaluations, 29 functions, ten seeds) it ranks first as well (1.97 versus 2.31 and 3.83 for the compared variants), winning 24 of 29 functions against standard GWO (ratio 0.721) and 24 of 29 against the frozen ANGWO reference (0.807). The sparse framework therefore scales upward in dimension and downward in budget.

D. Runtime

At equal budgets VAST-GWO is the most expensive method in wall-clock terms (5.93 s per run at $D = 50$/100,000 evaluations, versus 3.63 s for GWO, 3.41 s for HAGWO,

TABLE IV
EQUAL-SLOT CANDIDATE-SOURCE ABLATION (29 FUNCTIONS, 30 SEEDS, 100,000 TRUE EVALUATIONS, SEEDS 3071–3100). RATIOS ARE RELATIVE TO THE FULL CONFIGURATION; HOLM COUNTS SIGNIFICANT WINS/LOSSES OF THE LISTED COMPARISON.

Comparison	Med. ratio	W-L	Holm +/−	Po
Full vs A0 (GWO-only pool)	0.538	27-2	$+23/-0$	7×1
Full vs A1 (G1+G2)	0.812	25-4	$+21/-1$	$5\times$
Full vs A2 (G1+E)	0.994	14-15	$+1/-0$	
A2 vs A0	0.510	29-0	$+27/-0$	7×1
A5 (alpha-centred) vs Full	1.407	2-27	$+2/-23$	$1\times$

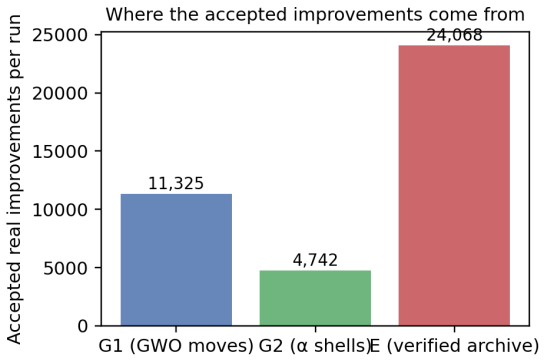


Fig. 3. Accepted real improvements per run by candidate route (mean over the formal ablation batch). The verified archive route dominates; the alpha-shell route is the smallest contributor, consistent with its optional role once the archive is present.

2.59 s for PSO, 2.41 s for CLPSO, 2.32 s for I-GWO and 4.54 s for AOGWO). This is the transparent price of building and maintaining a 100-candidate pool and a verified archive with a limited number of actual evaluations, and it is reported here so that practitioners can weigh quality against time.

VI. Ablation and Mechanism Analysis

A. Where the performance comes from

The ablation replaces disabled routes by additional standard GWO candidates so that the pool size ($M = 100$), the evaluated subset ($K = 10$) and the budget remain identical; the full configuration is bitwise identical to the frozen implementation on audited samples. Table IV reports the outcome. Removing both auxiliary routes (A0: GWO-only sparse pool) degrades the method to 27 losses in 29 functions (ratio 0.538, Holm $+23/-0$), which shows that sparse evaluation alone is not the source of the advantage. Removing the archive while keeping the alpha shells (A1) degrades the method to 25 losses (0.812, Holm $+21/-1$). Keeping the archive but removing the alpha shells (A2) leaves performance statistically unchanged (0.994, 14–15, pooled $p = 0.326$), i.e. the alpha-shell route is optional once the archive is present. Finally, replacing the archive by a single alpha-centred sampler with identical slots, scale and acceptance rule (A5) collapses the method to 2 wins in 29 functions (ratio 1.407, Holm $+2/-23$). The mechanism conclusion is therefore sharp: *the verified multi-region archive is the performance engine of VAST-GWO*, and the route structure, not the amount of sparse evaluation, is what must be engineered.

B. Source-level attribution

Instrumenting every commit with its route explains the ablation at the level of accepted moves: in a full run the archive route accounts for about 24,100 accepted real improvements per run on average, compared with about 11,300 for the GWO-move route and 4,700 for the alpha-shell route (Fig. 3). The archive route is thus not only present more often in the winning configuration; it is the route that actually converts evaluations into progress.

C. Design-space boundaries

The same controlled protocol was used to test the natural alternatives to uniform random allocation and to two popular enhancement strategies. Directed screening by incumbent geometry and history-informed selection were both worse than uniform sampling in development tests (ratios 1.39 and 1.53); fixed exploration quotas above the pool share degraded performance (up to 1.67); and three Q-learning quota schedulers never exceeded the best fixed policy (ratios 0.99–1.13) although they consistently beat random quota perturbation (0.75–0.87). A stagnation-triggered radius boost degraded performance (1.20). Finally, success-direction steering—extending moves along previously successful directions—improved final error against the frozen configuration (22–7, Holm $+2/-0$) but an equal-magnitude random-direction control achieved the same effect (ratio 0.988, pooled $p = 0.155$), and the gain appeared only in the last 20% of the budget, so the mechanism was rejected as an information-driven contribution. Together these boundaries support a simple design rule under scarce real evaluations: strengthen the *sources* of candidates and keep the allocation simple.

VII. Application: 3D UAV Delivery Path Planning

A. Scenario and protocol

The application is a three-dimensional UAV delivery problem built on five real Copernicus DEM terrain tiles (ridge, peaks, mountain, hills, plateau). Each instance contains 20 delivery stations with hard time windows, three cylindrical threat zones, and three intermediate control points per flight leg, giving a 209-dimensional mixed route-and-trajectory search space; the objective combines flight distance, waiting time and path smoothness under terrain-clearance, obstacle, altitude, climb-angle and turn-angle constraints. This is a normalized simulation derived from real terrain data, not a flight test. All methods run with $N = 50$, 30,000 true evaluations (initialization included), identical initial populations per (map, seed), and 30 seeds; 900/900 runs completed with zero failures.

B. Reliability under a censoring horizon

Table V reports reliability and quality. VAST-GWO is the only method that finds a feasible route in every one of its

TABLE V
3D UAV DELIVERY APPLICATION (150 RUNS PER METHOD). FEAS.: ROUTES FOUND WITHIN 30,000 EVALUATIONS; RMST: CENSORED MEAN TIME TO FIRST FEASIBILITY (LOWER IS BETTER); COSTS ARE MEDIANS OVER FEASIBLE RUNS.

Method	Feas. (%)	RMST	First feas.	Cost
VAST-GWO	**100.0**	**7062**	6314	667.4
GWO	94.0	10936	8920	699.8
I-GWO	97.3	12270	11152	724.7
HAGWO	97.3	8133	7432	663.4
PSO	95.3	8138	5357	766.4
CLPSO	61.3	24427	21330	1015.4

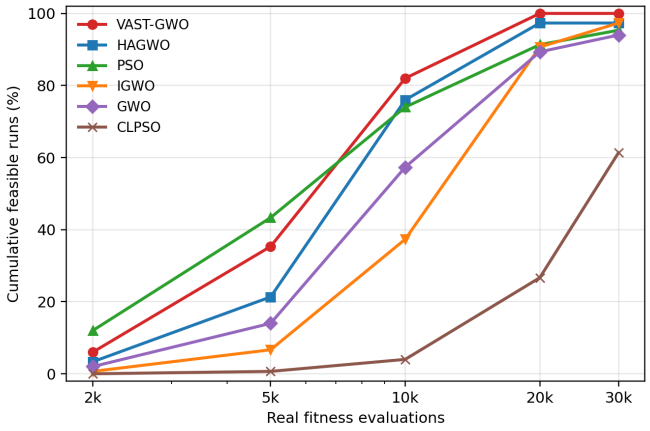


Fig. 4. Cumulative feasible runs versus real evaluations on the five DEM-based delivery scenarios (150 runs per method). Checkpoints are early observations within the single 30k protocol.

150 runs; the runner-up reliability is 97.3% (I-GWO and HAGWO), followed by PSO (95.3%), GWO (94.0%) and CLPSO (61.3%). Because unsuccessful runs would otherwise make median time-to-feasibility optimistic, we report the restricted mean time to first feasibility (RMST) with a 30,000-evaluation censoring horizon, in which every run—successful or not—enters the statistic. VAST-GWO attains the smallest RMST (7,062 evaluations, 95% CI 6,449–7,731), ahead of HAGWO (8,133) and PSO (8,138); the cumulative feasibility curves (Fig. 4) show the same ordering developing from the middle of the budget onwards, with 82.0% feasibility at 10,000 evaluations and 100% at 20,000 for VAST-GWO.

C. Route quality and paired tests

Final route quality favours VAST-GWO against every baseline after Holm correction over the five baselines: median paired ratios range from 0.631 (CLPSO) to 0.967 (HAGWO), with Holm p from 3.3×10^{-16} to 0.0458. The McNemar tests on final feasibility are significant against GWO, PSO and CLPSO; against I-GWO and HAGWO the observed reliability advantage is not significant with four discordant pairs, which we report for completeness. Two comparisons deserve explicit reading. First, PSO finds feasible routes earlier in the first few thousand evaluations (median 5,357 vs. 6,314), and its paired time-to-feasibility difference is not significant ($p = 0.598$ on the checkpoint-ordered test); in exchange, its final routes are

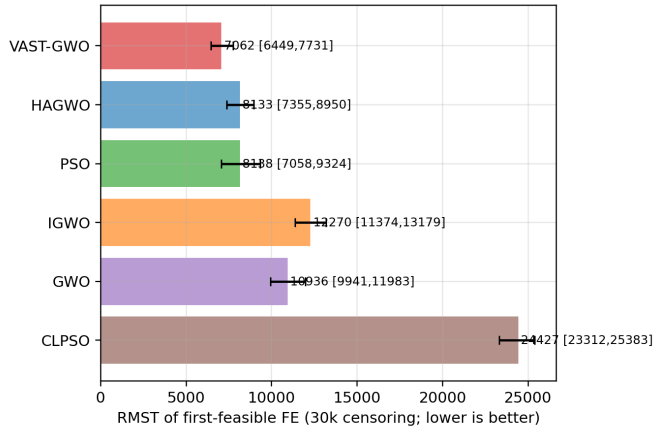


Fig. 5. Censored mean time to first feasible route (RMST, 30k horizon, bootstrap 95% CI). Every run enters the statistic.

markedly worse (median cost 766.4 vs. 667.4) and it misses feasibility in 7 of 150 runs. Second, HAGWO's marginal median cost (663.4) is slightly below that of VAST-GWO (667.4) because it conditions on its own successful runs; on the 146 pairs where both methods succeed, VAST-GWO is significantly better (ratio 0.967, rank-biserial 0.19, Holm $p = 0.0458$), and it is fully reliable. Fig. 6 shows the best route found by VAST-GWO on each of the five scenarios.

VIII. DISCUSSION

The experiments support a compact statement about optimizer design under scarce real evaluations: *the composition of the candidate pool is a first-class design variable, and it can outperform additional scheduling complexity*. In VAST-GWO this composition is a verified archive spanning several regions of the search space. Because every archived centre is backed by a billed evaluation, the archive can safely generate candidates far from the current leaders without introducing unverified optimism; because it spans multiple regions, it counteracts the leader-driven contraction that makes GWO stagnate on multimodal landscapes; and because the allocation remains a simple uniform draw, none of the gains depend on a fragile scheduler, as the boundary experiments confirm. The ablation makes the same point from the opposite direction: once the archive is present, the alpha-shell route is optional, and a single-centre substitute for the archive destroys performance—multi-region verified memory, not local refinement, is what the sparse budget needs.

Two properties recommend VAST-GWO for practice. First, its behaviour is robust across operating points: it ranks first both at $D = 100$ and under a budget five times smaller. Second, it transfers to a demanding constrained application without re-tuning: on the 209-dimensional UAV delivery problem it is the only method that is feasible in every run, and it combines this reliability with the best censored time-to-feasibility and the best final route quality against all baselines.

Scope and future work. The present study covers the CEC2017 family at $D = 50/D = 100$ plus one application

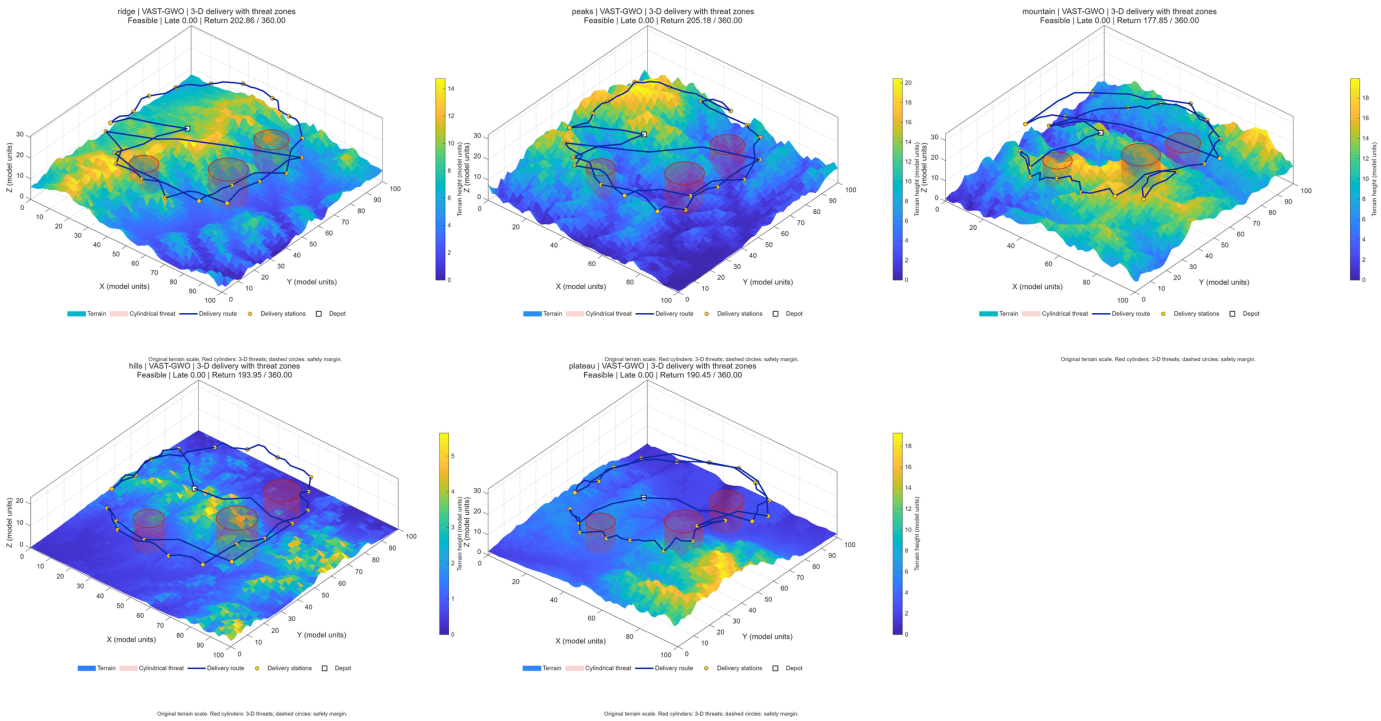


Fig. 6. Best feasible delivery routes found by VAST-GWO on the five DEM-based scenarios (costs 481.7–600.7). Terrain colour encodes elevation; red cylinders are threat zones with safety margins; the blue polyline is the route, dots are stations and the square is the depot.

family (terrain-constrained 3D delivery); extending the evidence to further engineering domains and to higher dimensions is a natural next step, and surrogate coupling—using the verified archive as training data for a lightweight model—is a promising direction that the verified-state discipline makes particularly clean. The ablation additionally indicates that a simplified variant that keeps only the GWO-move and archive routes is a worthwhile subject for future study, since the alpha-shell route was found optional in the presence of the archive.

IX. Conclusion

This paper introduced VAST-GWO, a verified archive-guided sparse-evaluation grey wolf optimizer for optimization under limited real-evaluation budgets. The method separates cheap candidate generation (100 candidates from GWO moves, success-scale alpha shells and a verified multi-region elite archive) from real evaluation (ten uniformly sampled candidates per generation), and admits only verified information into leaders, archives and rankings. On CEC2017 at $D = 50$ with 100,000 true evaluations and 30 seeds, VAST-GWO ranks first among seven methods, significantly outperforming five of six baselines including the 2026 hierarchical HAGWO and the 2026 adaptive oppositional AOGWO, while reaching statistical parity with the strongest classical improvement. Controlled equal-slot ablations show that the verified archive is the performance engine, and additional operating points ($D = 100$; 20,000 evaluations) confirm the ranking. In a 209-dimensional DEM-based 3D UAV delivery application with hard time windows, VAST-GWO is the only method feasible in all 150 runs and achieves both the best censored time-to-feasibility and the best final route quality against every baseline. The broader lesson is that under scarce real evaluations, engineering the sources of candidates—and verifying everything that enters memory—pays off more than engineering the scheduler.

Data and Code Availability

All optimization and application experiments, including raw per-run records, the experiment registry (every configuration and negative result), the ablation implementation, the UAV scenarios and the analysis scripts, are available in the accompanying repository; the evaluation counts of all runs are asserted by an independent probe, and the frozen method configuration is bitwise audited against its archived implementation.

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